

Evidence 1a

- [] **EC2 Instance:** Running Amazon Linux 2023 with the name tag `lab-ec2-app`.
- [] **RDS Instance:** MySQL engine, state = `available`, and **Public Access = No**.
- [] **Security Group (sg-rds-lab):** Inbound rule allows TCP 3306 **only** from the EC2 Security Group (`sg-ec2-lab`).
- [] **IAM Role:** Attached to EC2 with a policy allowing `secretsmanager:GetSecretValue`.
- [] **Application:** Screenshot of: RDS SG inbound rule using source = sg-ec2-lab EC2 role attached /list output showing at least 3 notes

- [] **EC2 Instance:** Running Amazon Linux 2023 with the name tag `lab-ec2-app`.

```
ap-northeast-1 +
~ $ aws ec2 describe-instances --filters "Name=tag:Name, Values=armageddon-lab-1a-ec2-app" --query "Reservations[].Instances[].State.Name"
[
  "running"
]
```

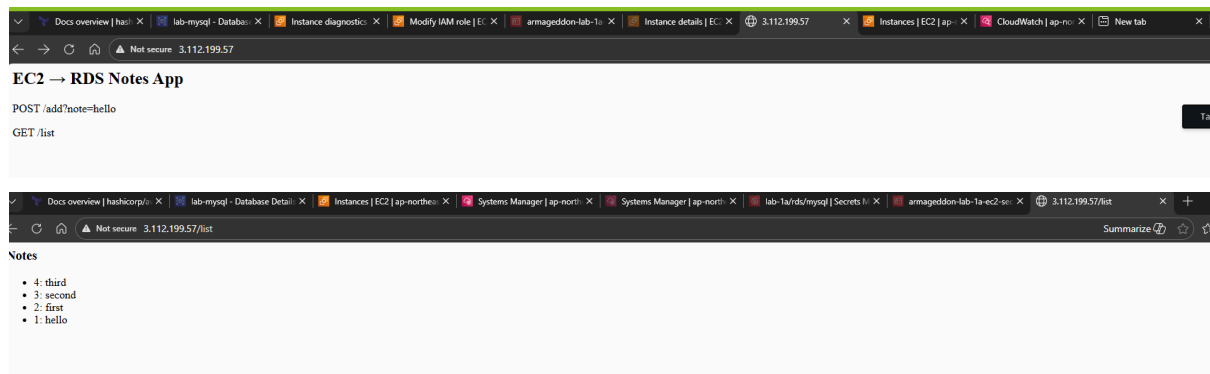
- [] **RDS Instance:** MySQL engine, state = `available`, and **Public Access = No**.

```
~ $ aws rds describe-db-instances --db-instance-identifier lab-mysql --query "DBInstances[].PubliclyAccessible"
[
  false
]
```

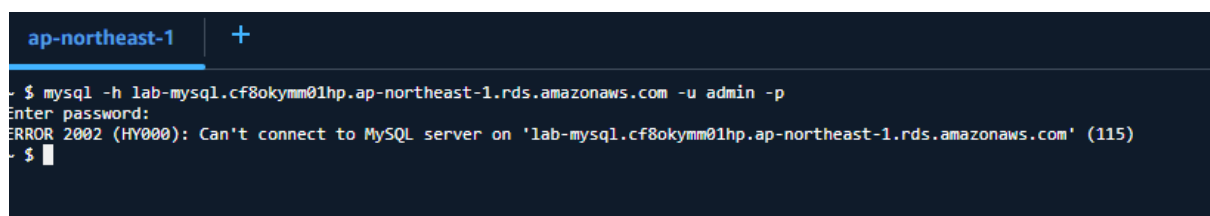
- [] **Secret Access:** `aws secretsmanager get-secret-value --secret-id lab/rds/mysql` (Run from EC2)

```
~ $ aws secretsmanager list-secrets
{
  "SecretList": [
    {
      "ARN": "arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab-1a/rds/mysql-PZkd1j",
      "Name": "lab-1a/rds/mysql",
      "LastChangedDate": "2026-01-29T20:17:16.220000+00:00",
      "LastAccessedDate": "2026-01-29T00:00:00+00:00",
      "SecretVersionsToStages": {
        "terraform-20260129193828090000000002": [
          "AWSPREVIOUS"
        ],
        "terraform-20260129201719030000000001": [
          "AWSCURRENT"
        ]
      },
      "CreateDate": "2026-01-16T23:06:33.909000+00:00"
    }
  ],
  "Truncated": false
}
```

- Application:** Screenshot of: RDS SG inbound rule using source = sg-ec2-lab EC2 role attached /list output showing at least 3 notes



Database connection



Reflection Questions (Lab 1A)

- **A)** Why is DB inbound source restricted to the EC2 security group instead of an IP range? *This is to limit access (apply the principle of least privilege). Using the IP makes you vulnerable to attacks*
- **B)** What port does MySQL use by default? 3360
- **C)** Why is Secrets Manager superior to storing credentials in code or **user-data**? Secrets manager comes with encryption and password rotation after a predetermined set of time. This enhances security for the application.